

## Practitioner's Commentary: The Outstanding Railroad Flatcar Papers

John J. Bartholdi, III

School of Industrial and Systems Engineering  
Georgia Institute of Technology  
Atlanta, GA 30332

*"It is difficult  
to be critical  
of these  
papers...."*

It is difficult to be critical of these papers; after all, each one solved the problem! That is, each team found 30 or so ways of loading the railroad cars with only 6 mm of unused space, and proved that no other arrangement could do better. Each group recognized that the problem is easy in the sense that there are "only" finitely many ways of loading the cars; thus the problem is in principle solvable by enumerating all possibilities and choosing a best way. The difficulty, of course, is that this technique can be hopelessly impractical. There are two ways out of this impasse: one is to rely on special structure in the problem to reduce the number of possibilities to be considered; the other is to rely on an efficient procedure that produces acceptably good though possibly suboptimal solutions. All four groups took the first approach.

The problem is of a type known as a "1-dimensional packing": how to efficiently pack 1-dimensional "rectangles" (segments) into a larger 1-dimensional "rectangle" (segment). There is an active literature analyzing the performance of heuristics for such problems. It is all quite interesting, especially because simple heuristics can have quite surprising—and sometimes quite difficult to prove—performance guarantees. (For classic results see [Johnson et al. 1974]; for some tantalizing empirical results, see [Bentley et al. 1983].) The activity in this area has been spreading to include analysis of the average-case performance of heuristics (for example, [Coffman et al. 1988]), and 2-dimensional packing, wherein we must pack smaller (2-dimensional) rectangles into a larger one. Naturally, some brave souls have ventured forth to pack  $n$ -dimensional rectangles. (A particularly interesting example, and some fine mathematics, can be found in [Karp et al. 1984].) Of course, the rail car problem is actually a bit more than the typical 1-dimensional packing problem, since there is, in addition to the decision of what gets packed, the decision of on which rail car to pack it.

I liked the way the University of Toronto team began by estimating the computational requirements of two very general methods of enumeration, brute-force and dynamic programming. I think it is good engineering sense to start by estimating the size of the search space; after all, sometimes simply throwing computing cycles at a problem is appropriate, especially if there is little structure to be exploited; the possibility should at least be considered, even if such solutions are not completely satisfying.

*"All of the teams relied on exploiting structure...."*

All of the teams recognized the impracticality of uninformed, brute-force search; they relied on exploiting structure special to this instance. All used primal techniques that generated feasible packings; thus, if too time-consuming, the search could be halted with the best solution found so far. To their credit, the members of the team from the US Military Academy looked beyond this particular instance by implementing a general solution procedure. Their approach (branch-and-bound) is widely applicable and generally effective; it would probably work adequately for all but the largest problems. (It should be noted that "Krolak's bounded-variable method" has nothing particular to recommend it; it is just one of many ways of performing a branch-and-bound search.)

The Harvard University team also suggested a general approach: that one balance by enumeration a "composite" car that is twice as long, and then partition the selected boxes by enumeration. I am dubious; this requires two enumerations, both of which are potentially time-consuming, and there is no guarantee that an effective packing for a composite car can be partitioned into two good packings for individual cars.

*"Any packing that fits so tightly is in danger of being unusable."*

*"... the better the solution to the model, the less likely it is to be feasible...."*

Since the suggested packings of the rail cars wasted so little space (6 mm), it is important to do sensitivity analysis. Any packing that fits so tightly is in danger of being unusable. One must have a lot of confidence in the data to recommend packing two rail cars to within so small a tolerance—one does not want the freight loaders doing a feasibility check with fork lifts! It seems to me that for a real problem the length of the rail car is likely to be known quite accurately, but the dimensions of the boxes are likely to be somewhat inaccurate. This might be because of hasty or careless measurement by freight dispatchers, or because the boxes are not perfectly square due to imprecise construction, or bulging, or protruding flanges, hinges, or braces. Because of this uncertainty, we have the dilemma that the better the solution to the model, the less likely it is to be actually feasible to the real problem!

The University of Toronto team did a very thorough deterministic sensitivity analysis. They observed that, as seems frequently to be true of packing problems, the solutions were quite sensitive to the lengths of the boxes to be packed. This was especially so here, since

*"...reason to worry that we have solved the model but not the problem!"*

the optimum packing wasted so little space. They observe that if each box was actually just 0.17 mm longer than its reported length<sup>13</sup>, then the optimal packing is no longer feasible—all the more reason to worry that we have solved the model but not the problem!

While a deterministic sensitivity analysis can be useful, sometimes it is too expensive computationally. It seems to be a feature of difficult combinatorial problems that the solutions are "fragile," in the sense that small perturbations can destroy feasibility or optimality. Furthermore, there is a certain "turbulence" among solutions: small changes in the data make large changes in the solutions. To make matters worse, it is typically the case that performing sensitivity analysis is as difficult as solving the problem, so that we must resort to other means.

An alternative approach was suggested by the US Military Academy team, who suggested the appropriateness of a probabilistic model of error. However, it would have been nicer had they made their model of measurement error explicit. For example, one might assume that the error of each measurement is random according to some distribution over  $(-0.5, +0.5)$ ; then for each packing one could compute the probability that it will actually fit on the cars.

To complement its solution of the problem, the Harvard University team showed that a natural generalization of the problem is NP-complete. While not a surprise, this can be interpreted as justifying the search for effective heuristics. They suggest one: generate a random packing and repeatedly perturb it at random, retaining the best. Such heuristics are appealing; they are generally quite simple to implement, and they can be modified to apply to many types of combinatorial problems. Unfortunately, experience so far suggests that such approaches are not particularly effective unless there is some way of guiding the random search to exploit problem structure. Again, as so often seems to be the case in combinatorial problems, special structure is the key to effective solution; it can be ignored only at potentially great computational cost.

Now a word about actually computing the solutions. I found it interesting that the teams programmed in FORTRAN, APL, and Turing (new to me, but identified as a variant of Pascal). I cannot help it: I am always surprised when I find someone programming in FORTRAN when they do not have to! I found the APL remarkably concise but remarkably cryptic; it can be a useful tool, but is probably not a good language for expressing algorithmic ideas to others. Judged by this standard, most of these programs were written at a quite low level of abstraction: there were no abstract data types called "Box" or "Rail\_Car", no subroutine called "Test\_Fit", or other natural data or procedural abstractions. As was to be expected, the teams were more focussed on finding a solution to this problem than on writing a general rail-car packer.

*"...most of these programs were written at a quite low level of abstraction..."*

*"...this is  
a recurring  
problem...."*

Each of the groups did quite well (and are encouraged to apply to graduate school at Georgia Tech!). I am hard-pressed to lecture them about anything. Instead, let me suggest another way of interpreting the problem that emphasizes the modeling rather than the mathematics. It seems likely to me that this is a recurring problem; indeed, freight depots must make similar decisions daily about what gets loaded where. When interpreted this broadly, the problem—and what counts as a solution—looks somewhat different. Now it seems less important to produce any one good packing (after all, how important can it be to waste a few centimeters of rail car space?); instead what is wanted is a fast method to produce acceptably good solutions. To tackle this problem, one would want to take the same general approach as did the four groups: exploit special structure. However, now it seems more important to exploit structure of the general problem rather than that of any particular instance. (We cannot depend on conveniences like the trucking constraint, which here reduced the search space by several orders of magnitude.)

Let us speculate about the "real" problem:

*"...the boxes  
that are not  
loaded in this  
problem must  
nevertheless be  
sent!"*

- The dimensions of the boxes are unlikely to vary by more than a few (1?) orders of magnitude. We will probably not be loading both diesel generators and thimbles. This means that there will be many packings of approximately the same quality; it will be easier to find a decent packing, but harder to find the best, since the terrain of the solution space looks approximately the same in all directions. This suggests the appropriateness of a fast heuristic.

- Presumably the boxes that are not loaded in this problem must nevertheless be sent! If we must rent additional rail cars for them, then all that matters is how many additional cars are required. Whether the first car is packed to within 1 or 1000 mm does not matter: a third car will be required. This suggests that it is not so important to reduce wasted space as it is to reduce the number of rail cars required to transport the boxes. In fact, the objective "wasted space" looks to me like a surrogate for the objective "number of rail cars."

- I suspect that the larger problem is that both boxes and trains arrive intermittently at a depot; we want to load the boxes onto the trains in some efficient manner. I imagine at least two conflicting goals: to avoid wasting space on the train, and to avoid having boxes wait excessively long to be shipped. We can improve our packing if we delay some boxes until others arrive that fit better with them, but this might delay some boxes unacceptably; if we load boxes in batches according to the order in which they arrive, then we will not do as well in packing since we restrict our options.

Finally, besides having written a good paper, the UC-Berkeley team had an inspired title!